WORKED SOLUTION

STEP I 1998, Question 9

Mechanics • Collisions of Pendulums

STEP I 1998, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q9

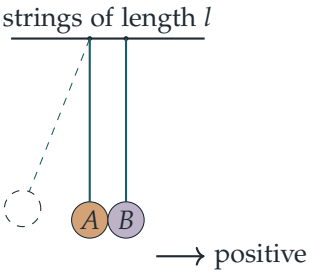
The key idea

Because the displacement is very small, each sphere moves as a simple pendulum in simple harmonic motion, and the period $2\pi\sqrt{l/g}$ doesn't depend on the amplitude or the speed. The two pendulums have the same length, so after any collision at the lowest point both spheres come back to the lowest point at the same moment, half a period later, with their velocities reversed. Once we know where and when the second impact happens, it's two lines of momentum and Newton's experimental law each time.

Solution

The model

We treat A and B as particles of mass m on light inextensible strings of the same length l, ignore air resistance, and take the collisions as instantaneous, with impulses along the horizontal line of centres. Take the positive direction to be the direction A is moving at the first impact (from A towards B), and let $\omega = \sqrt{g/l}$.



Why every collision is at the lowest point

For small oscillations each sphere moves with simple harmonic motion of angular frequency ω, whatever its amplitude. So a sphere that leaves the lowest point with

speed v has horizontal displacement $\frac{v}{\omega}\sin\omega t$, and it is back at the lowest point, moving with velocity $-v$, after a time π/ω (half a period) — for every v.

The first impact is at the lowest point: B hangs there at rest, and A reaches it with speed u. Suppose a collision happens at the lowest point and afterwards A and B have velocities $p \le q$ (they can't move towards each other just after impact). Measured from the lowest point, the gap between the spheres at time t later is

$$\frac{q}{\omega}\sin\omega t - \frac{p}{\omega}\sin\omega t = \frac{q-p}{\omega}\sin\omega t,$$

which is positive for $0 < t < \pi/\omega$ and zero at $t = \pi/\omega$. So they can't meet earlier, and they meet again exactly when both are back at the lowest point. That's the hint: the two pendulums have the same period, so they always return to the lowest point together. By induction, every collision is at the lowest point, and consecutive collisions are half a period apart.

The first impact

Let A and B have velocities v_A and v_B just after the first impact. Conservation of momentum (the string tensions are vertical at the lowest point, so they give no horizontal impulse) and Newton's experimental law give

$$\begin{aligned} mv_A + mv_B &= mu, \\ v_B - v_A &= eu. \end{aligned}$$

Adding and subtracting,

$$v_A = \tfrac{1}{2}u(1-e), \qquad v_B = \tfrac{1}{2}u(1+e).$$

Both are ≥ 0, so both spheres swing out on B's side (if $e = 1$, A stops dead and stays hanging at the lowest point).

Ask yourself What are the velocities just before the second impact?

Half a period later each sphere is back at the lowest point with its velocity reversed: A has velocity $-\frac{1}{2}u(1-e)$ and B has velocity $-\frac{1}{2}u(1+e)$. B is the faster, so it catches A from behind. Its speed of approach relative to A is $\frac{1}{2}u(1+e) - \frac{1}{2}u(1-e) = eu$.

The second impact

Let A and B have velocities w_A and w_B just after the second impact. Conservation of momentum gives

$$mw_A + mw_B = -\tfrac{1}{2}mu(1-e) - \tfrac{1}{2}mu(1+e) = -mu,$$

and Newton's experimental law (the speed of separation is e times the speed of approach, eu) gives

$$w_B - w_A = e \cdot eu = e^2u.$$

Subtracting and adding,

$$w_A = -\tfrac{1}{2}u(1+e^2), \qquad w_B = -\tfrac{1}{2}u(1-e^2).$$

So immediately after the second impact A has speed $\frac{1}{2}u(1+e^2)$, as required, and B has speed $\frac{1}{2}u(1-e^2)$, both moving towards the side from which A was originally released.

Checks

- Perfectly elastic, $e = 1$: A ends with speed u and B is at rest. That's right: after the first impact A stops and B takes speed u, and B returns with speed u and gives it all back to A.
- $e = 0$: both speeds are $\frac{1}{2}u$. The spheres coalesce in effect, moving together with speed $\frac{1}{2}u$ after the first impact, so this is right too.
- Momentum after the second impact is $-\frac{1}{2}mu(1+e^2) - \frac{1}{2}mu(1-e^2) = -mu$, the reverse of the original mu.
- We simulated two real (nonlinear) pendulums with $l = 1\,\text{m}$ and an initial angle of 0.01 rad. The second impact came 1.0001 half-periods after the first, within 3×10^{-6} rad of the lowest point, and with $e = 0.7$ the speeds after it were $0.745u$ and $0.255u$, matching $\frac{1}{2}(1+e^2)$ and $\frac{1}{2}(1-e^2)$.

Common slips

- Forgetting that both velocities are reversed before the second impact, and treating B as hitting a stationary A.
- Using eu in place of $e \cdot (\text{speed of approach})$ at the second impact. Here the approach speed happens to be eu, so the separation speed is e^2u.
- Saying the collisions are at the lowest point "because that is where the spheres hang". The real reason is that the period doesn't depend on the amplitude, and that's why the drawing aside has to be small.

Beyond STEP

Isochronism (a period that doesn't depend on amplitude) holds only approximately for a pendulum: for larger swings the period grows, given exactly by an elliptic integral, and the collisions would drift away from the lowest point. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both the pendulum and collisions with care.

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